Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part c.ii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (a) | A1<br>A2<br>A3<br>A4 | 2.1.4 | Award 1 mark for each appropriate validation test (up to 2) and 1 mark for suitable example of erroneous data that matches the test e.g.<br><br>First 3 characters are upper case letters: Fff123456o<br>Characters 4 to 9 are non-zero numbers: FFF101111o<br>Consists of only 10 characters: FFF111111ee<br>Final character correct for sum of numbers: FFF112233o | | (4) |

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Functions, procedures, parameters and libraries

### Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part c.ii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (b) (i) | B1<br>B2 | 1.1.9 | Award 1 mark each for any of:<br>• Simple implementation (1)<br>• Can be used for sorted or unsorted lists (1)<br>• If the target is at the beginning of the data structure the search will be faster than a binary search | Accept disadvantages of a binary compared with a linear search. | (2) |

Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part c.ii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (b) (ii) | B3<br>B4 | 1.1.9 | Award 1 mark each for any of:<br>• May need to compare with all items in list before search complete (1)<br>• May need longer time for searching a large list (1) | Accept advantages of a binary compared with a linear search. | (2) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 4 of 7 · find part c.ii in this mark scheme extract

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Functions, procedures, parameters and libraries

### Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part c.ii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (c) (i) | C1<br>C2 | 2.6.2 | Award 1 mark each up to a maximum of 2 for:<br>• Both can use arguments / parameter passing (1)<br>• Both can use local variables (1)<br>• Both can be called from anywhere within the program (1)<br>• Code can be reused without being rewritten (1)<br>• Both can be library files (1)<br>• The code can be independently tested for both (1) | | (2) |

Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 6 of 7 · find part c.ii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (c) (ii) | C3<br>C4 | 2.6.2 | Award two marks for a linked explanation such as:<br><br>• A function must always return a result (1) whereas a procedure does not (1)<br>• A function interface must have a data type (1) to signify the type of data that will be returned / a procedure does not need this (1)<br>• The result of a function must always be used (either assigned to a variable or as part of a condition) (1) whereas a procedure does not explicitly return a result to be used (1)<br>• A function produces information (1) whereas a procedure performs a task (1) | | (2) |

8 of 16

Type MS · Functions, procedures, parameters and libraries

Answer 1 of 6 · Question 5 part c.ii · 2 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part c.ii in this mark scheme extract

**Mark Scheme – Python Coding**

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

Type MS · Functions, procedures, parameters and libraries

Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part c.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (a) | A1<br>A2<br>A3<br>A4 | 2.1.4 | Award 1 mark for each appropriate validation test (up to 2) and 1 mark for suitable example of erroneous data that matches the test e.g.<br><br>First 3 characters are upper case letters: Fff123456o<br>Characters 4 to 9 are non-zero numbers: FFF101111o<br>Consists of only 10 characters: FFF111111ee<br>Final character correct for sum of numbers: FFF112233o | | (4) |

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

Type MS · Functions, procedures, parameters and libraries

Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part c.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (b) (i) | B1<br>B2 | 1.1.9 | Award 1 mark each for any of:<br>• Simple implementation (1)<br>• Can be used for sorted or unsorted lists (1)<br>• If the target is at the beginning of the data structure the search will be faster than a binary search | Accept disadvantages of a binary compared with a linear search. | (2) |

Type MS · Functions, procedures, parameters and libraries

Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part c.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (b) (ii) | B3<br>B4 | 1.1.9 | Award 1 mark each for any of:<br>• May need to compare with all items in list before search complete (1)<br>• May need longer time for searching a large list (1) | Accept advantages of a binary compared with a linear search. | (2) |

7 of 16

## Type MS · Functions, procedures, parameters and libraries

### Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 4 of 7 · find part c.i in this mark scheme extract

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Functions, procedures, parameters and libraries

### Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part c.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 5 (c) (i) | C1<br>C2 | 2.6.2 | Award 1 mark each up to a maximum of 2 for:<br>• Both can use arguments / parameter passing (1)<br>• Both can use local variables (1)<br>• Both can be called from anywhere within the program (1)<br>• Code can be reused without being rewritten (1)<br>• Both can be library files (1)<br>• The code can be independently tested for both (1) | | (2) |

| | | | | | |
|---|---|---|---|---|---|
| 5 (c) (ii) | C3<br>C4 | 2.6.2 | Award two marks for a linked explanation such as:<br><br>• A function must always return a result (1) whereas a procedure does not (1)<br>• A function interface must have a data type (1) to signify the type of data that will be returned / a procedure does not need this (1)<br>• The result of a function must always be used (either assigned to a variable or as part of a condition) (1) whereas a procedure does not explicitly return a result to be used (1)<br>• A function produces information (1) whereas a procedure performs a task (1) | | (2) |

Type MS · Functions, procedures, parameters and libraries

Answer 2 of 6 · Question 5 part c.i · 2 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part c.i in this mark scheme extract

**Mark Scheme – Python Coding**

| Question | mp | ref | Answer | Additional Guidance | Mark |
|---|---|---|---|---|---|

## Type MS · Functions, procedures, parameters and libraries

### Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 1 of 8 · find part d.iii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (b) | B1 | Award up to 2 marks for a linked explanation such as:<br>• The number of keys are limited/only one shift used (1) making it easy to use brute force to decrypt (1)<br>• It can be easy to find commonly used letters (e.g. E) (1) and guess the key (1) | Accept alternative similar wording. | (2) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 2 of 8 · find part d.iii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (c) | | Award 1 mark each up to a maximum of 4 for:<br><table><tr><td>Encrypted letter</td><td>f</td><td>l</td><td>m</td><td>k</td><td>t</td><td>r</td><td>w</td><td>h</td><td>e</td><td>e</td></tr><tr><td>Keyword letter</td><td>t</td><td>h</td><td>i</td><td>r</td><td>t</td><td>y</td><td>t</td><td>h</td><td>i</td><td>r</td></tr><tr><td>Decrypted letter</td><td>m</td><td>e</td><td>e</td><td>t</td><td>a</td><td>t</td><td>d</td><td>a</td><td>w</td><td>n</td></tr></table> | | |
| | C1 | Award 1 mark from:<br>• flmktrwhee mapped to thirtythir (1)<br>• Flmktr mapped to thirty (1) | | |
| | C2 | At least one letter decrypted correctly (1) | | |
| | C3 | At least one word decrypted correctly (1) | | |
| | C4 | Decrypted message 'meet at dawn' (1) | | (4) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 3 of 8 · find part d.iii in this mark scheme extract

| Question | IMP | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (d)(i) | D1 | Award 1 mark for:<br>• cipher/ letter / a single encrypted letter (1) | Do not accept word/message/text | (1) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 4 of 8 · find part d.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | | • cipherLetter / a single encrypted letter (1) | | (1) |
| 3 (d)(ii) | D2 | Award 1 mark for any of:<br>• keywordLetter<br>• plaintextLetter | Ignore case | (1) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 5 of 8 · find part d.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | | • plaintextLetter | | (1) |
| 3 (d)(iii) | D2 | Award 1 mark for any of:<br>• subprogram that is already defined<br>• subprogram that is already written<br>• subprogram that is already compiled<br>• subprogram that can be called without having to write code for it | Must be clear they are referring to subprogram provided by the language itself | (1) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 6 of 8 · find part d.iii in this mark scheme extract

## Type MS · Functions, procedures, parameters and libraries

### Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 7 of 8 · find part d.iii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (a) | A1 | Get plaintext and store in plaintext variable | Accept alternative wording<br>Line numbers may be different compared to the examples shown<br><br>When testing the completed code use lowercase for the input | (1) |
| | A2 | Get key and store in key variable | | (1) |
| | A3 | Validate key (must repeat until a valid key has been input) | | (1) |
| | A4 | Open file to write/append | | (1) |
| | A5 | Write cipher text | | (1) |
| | A6 | Close file | | (1) |
| | A7 | Displays ciphertext | | (1) |
| | A8 | Executing and producing correct output to file and screen (must have A3) | | (1) |
| Code examples | | | | |
| Python | | | | |

```
15
16 #       Add your code to get the plaintext and convert it to lowercase
17 plaintext = input("Enter the plaintext using lowercase letters: ")
18 plaintext = plaintext.lower()
```

```
# Add your code to get the key and make sure the key is between 1 and 25
while key < 1 or key > 25:
    key = int(input("Enter the key - a number between 1 and 25 "))
```

```
39 # Add your code to write the ciphertext to a text file
40 cipherFile = open("Cipher.txt", "w")
41 cipherFile.write(ciphertext)
42 cipherFile.close()
```

```
40
41 #       Add your code to display the ciphertext
42 print("The ciphertext message is:",ciphertext)
43
```

## Type MS · Functions, procedures, parameters and libraries

### Answer 3 of 6 · Question 3 part d.iii · 1 mark

2021 JUN · Paper 02 · extract 8 of 8 · find part d.iii in this mark scheme extract

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 1 of 11 · find part b in this mark scheme extract

| Question | | Answer | Additional Guidance | Mark |
|---|---|---|---|---|
| 3 (a) (i) | A1 | C# and Python monthNumber/monthsAndDays/count/found (1)<br>Java monthNumber/monthsAndDays/count/found/input | | (1) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 2 of 11 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (ii) | A2 | monthsAndDays (1) | (1) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 3 of 11 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iii) | A3 | C#<br>• WriteLine, ToString, ToInt32, ReadLine<br>Java<br>• Print, ParseInt, nextLine, ToString, Equals<br>Python<br>• print/len/int/input (1) | (1) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 4 of 11 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (iv) | A4 | Display/showMonthNameAndDays (1) | (1) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 5 of 11 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (v) | A5 | pMonth (1) | (1) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 6 of 11 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) (vi) | A6<br>A7 | **Award 1 mark for each of:**<br><br>**Type of test data** / **Test data**<br>Boundary (1) / 1<br>Erroneous (1) / 15 | | (2) |

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 7 of 11 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (b) | B1<br>B2 | **Award up to 2 marks for a linked explanation such as:**<br><br>• A library program would not know the names of the calling program's variables in advance (1) meaning it would not be a reusable solution (1)<br>• (Using parameters) enables reusable solutions (1) using different data values/arguments (1)<br>• Using parameters uses less memory / uses memory more efficiently (1) because it avoids the use of global variables / because the memory is freed after the subprogram is finished (1) | | (2) |

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

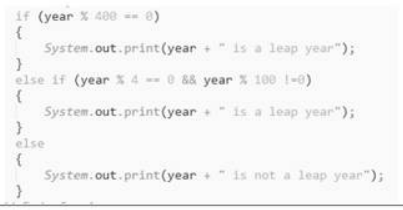
2021 NOV · Paper 02 · extract 8 of 11 · find part b in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 9 of 11 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (c) | **Award 1 mark for each of:** | | | |
| | C1 | Condition to check whether the year exactly divides by 4 or equivalent condition (1) | | |
| | C2 | Condition to check whether year exactly divides by 400 or equivalent condition (1) | | |
| | C3 | Condition to check whether year does not exactly divide by 100 or equivalent condition (1) | | |
| | C4 | Correct message for at least one of the conditions (1) | | |
| | C5 | Program executes and produces the correct output (1) | | |
| | C6 | Efficient solution (1) | | |
| | | | | (6) |
| Code examples | | | | |
| C# |  | | | |
| Java |  | | | |

```
if (year % 400 == 0)
{
    Console.WriteLine(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    Console.WriteLine(year + " is a leap year");
}
else
{
    Console.WriteLine(year + " is not a leap year");
}
```

```
if (year % 400 == 0)
{
    System.out.print(year + " is a leap year");
}
else if (year % 4 == 0 && year % 100 !=0)
{
    System.out.print(year + " is a leap year");
}
else
{
    System.out.print(year + " is not a leap year");
}
```

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 10 of 11 · find part b in this mark scheme extract

| | |
|---|---|
| Python | `if year % 400 == 0:`<br>`  print(year,"is a leap year")`<br>`elif year % 4 == 0 and year % 100 !=0:`<br>`  print(year,"is a lear year")`<br>`else:`<br>`  print(year,"is not a leap year")` |

## Type MS · Functions, procedures, parameters and libraries

### Answer 4 of 6 · Question 3 part b · 2 marks

2021 NOV · Paper 02 · extract 11 of 11 · find part b in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Functions, procedures, parameters and libraries

### Answer 5 of 6 · Question 2 part a.i · 2 marks

2023 JUN · Paper 02 · extract 1 of 4 · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2(a)(i) | A1<br>A2 | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>• The code in the library has already been debugged/tested/optimised / library code less likely to have errors (1)<br>• May include specialised/proprietary functions (1)<br>• Library subprograms can be imported into code when needed / don't need to write code themselves (to save a programmer time/work) (1)<br>• making it more readable / makes code easier to understand (Code files become shorter) | Allow characteristics of a specific library (math, random), if it can be attributed to unique bullets<br><br>Don't allow quicker/easier/better without a reason being given<br><br>Accept answers about subprograms as candidate may have written these themselves<br><br>Line numbers are for guidance – mark answer as a whole | **2** |
| 2(a)(ii) | A3 | **The only correct answer is A**<br><br>***B*** *is not correct because iteration is not callable and does not return a result.*<br>***C*** *is not correct because a procedure, while callable, does not return a result.*<br>***D*** *is not correct because selection is not callable and does not return a result.* | | **1** |
| 2(a)(iii) | A4 | Award **one** mark for:<br><br>• A variable that exists/is accessible only in the subprogram/scope in which it is created (1) | | **1** |

## Type MS · Functions, procedures, parameters and libraries

### Answer 5 of 6 · Question 2 part a.i · 2 marks

2023 JUN · Paper 02 · extract 2 of 4 · find part a.i in this mark scheme extract

## Type MS · Functions, procedures, parameters and libraries

### Answer 5 of 6 · Question 2 part a.i · 2 marks

2023 JUN · Paper 02 · extract 3 of 4 · find part a.i in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(b)(i)</td><td>B1<br>B2<br>B3</td><td>Award one mark for each correct row.<br><table><tr><th></th><th>Adult</th><th>Children</th></tr><tr><td>Normal</td><td>1/2/3/4</td><td>1/2/3/4/5/6</td></tr><tr><td>Boundary</td><td>1/4</td><td>1/6</td></tr><tr><td>Erroneous</td><td>Any value <1 or >4</td><td>Any value <1 or >6</td></tr></table></td><td>Accept all valid values for normal and boundary<br>Do not accept written numbers (two) as question asks for ‘numeric data’ – this is testing range, not data types.</td><td>3</td></tr>
<tr><td rowspan="5">2(b)(ii)</td><td colspan="2">Award one mark for each of:</td><td rowspan="5">Runtime error (B7) could be fixed before logic error (B6)<br>Applies to all languages<br>Accept < for <=<br>Allow /-1000 as alternative to B6<br>/1000 could happen after initial location e.g. print(“Order”, weightNeed/1000, “kilograms)</td><td rowspan="5">4</td></tr>
<tr><td>B4</td><td>Change =+= to +</td></tr>
<tr><td>B5</td><td>Change > to <=</td></tr>
<tr><td>B6</td><td>Change order of subtraction to: requiredWeight - inStock</td></tr>
<tr><td>B7</td><td>Change 0 to 1000</td></tr>
</table>

## Type MS · Functions, procedures, parameters and libraries

### Answer 5 of 6 · Question 2 part a.i · 2 marks

2023 JUN · Paper 02 · extract 4 of 4 · find part a.i in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|

## Type MS · Functions, procedures, parameters and libraries

### Answer 6 of 6 · Question 2 part a.ii · 1 mark

2023 JUN · Paper 02 · extract 1 of 4 · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2(a)(i) | A1<br>A2 | Award **one** mark for any of the following up to a maximum of **two** marks:<br><br>• The code in the library has already been debugged/tested/optimised / library code less likely to have errors (1)<br>• May include specialised/proprietary functions (1)<br>• Library subprograms can be imported into code when needed / don't need to write code themselves (to save a programmer time/work) (1)<br>• making it more readable / makes code easier to understand (Code files become shorter) | Allow characteristics of a specific library (math, random), if it can be attributed to unique bullets<br><br>Don't allow quicker/easier/better without a reason being given<br><br>Accept answers about subprograms as candidate may have written these themselves<br><br>Line numbers are for guidance – mark answer as a whole | **2** |
| 2(a)(ii) | A3 | **The only correct answer is A**<br><br>***B*** *is not correct because iteration is not callable and does not return a result.*<br>***C*** *is not correct because a procedure, while callable, does not return a result.*<br>***D*** *is not correct because selection is not callable and does not return a result.* | | **1** |
| 2(a)(iii) | A4 | Award **one** mark for:<br><br>• A variable that exists/is accessible only in the subprogram/scope in which it is created (1) | | **1** |

## Type MS · Functions, procedures, parameters and libraries

### Answer 6 of 6 · Question 2 part a.ii · 1 mark

2023 JUN · Paper 02 · extract 2 of 4 · find part a.ii in this mark scheme extract

## Type MS · Functions, procedures, parameters and libraries

### Answer 6 of 6 · Question 2 part a.ii · 1 mark

2023 JUN · Paper 02 · extract 3 of 4 · find part a.ii in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(b)(i)</td><td>B1<br>B2<br>B3</td><td>Award one mark for each correct row.<br><table><tr><th></th><th>Adult</th><th>Children</th></tr><tr><td>Normal</td><td>1/2/3/4</td><td>1/2/3/4/5/6</td></tr><tr><td>Boundary</td><td>1/4</td><td>1/6</td></tr><tr><td>Erroneous</td><td>Any value <1 or >4</td><td>Any value <1 or >6</td></tr></table></td><td>Accept all valid values for normal and boundary<br>Do not accept written numbers (two) as question asks for ‘numeric data’ – this is testing range, not data types.</td><td>3</td></tr>
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<tr><td>B4</td><td>Change =+= to +</td></tr>
<tr><td>B5</td><td>Change > to <=</td></tr>
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</table>

## Type MS · Functions, procedures, parameters and libraries

### Answer 6 of 6 · Question 2 part a.ii · 1 mark

2023 JUN · Paper 02 · extract 4 of 4 · find part a.ii in this mark scheme extract

| Question | mp | Answer | Additional Guidance | Mark |
|---|---|---|---|---|